



NeuroFleetX: AI-Driven Urban Mobility Optimization System

An intelligent platform that transforms fleet operations through real-time visibility, AI-powered routing, and predictive maintenance for urban transportation networks.

Transforming Urban Fleet Operations

Urban fleet operations face critical challenges that impact efficiency, customer satisfaction, and operational costs. Traditional manual systems lack the intelligence needed for modern mobility demands.

Manual Booking Inefficiency

Legacy systems create booking delays and poor user experience without real-time vehicle availability visibility.

Routing Challenges

Without intelligent routing, fleets experience increased travel time, fuel costs, and missed delivery windows.

Predictive Gaps

Absence of predictive maintenance leads to unexpected breakdowns and poor coordination among stakeholders.



Core System Modules

A comprehensive solution built around six interconnected modules that deliver end-to-end fleet management capabilities.



Authentication & Role Management

Secure JWT-based access with role-based control for customers, drivers, and administrators.



AI Route Optimization

Intelligent routing that balances speed, efficiency, and traffic conditions for optimal travel.



Booking & Scheduling

Automated vehicle assignment with lifecycle management from booking to completion.



Fleet Inventory Management

Real-time vehicle status tracking with location, fuel levels, and operational metrics.



Predictive Maintenance

Real-time telemetry monitoring with AI thresholds that predict maintenance needs before breakdowns.



Analytics Dashboards

Role-specific dashboards providing real-time insights and data-driven decision support.



System Architecture

A scalable, modular architecture built with modern web technologies that ensure reliability and performance.

01

Front-End Layer

React-based single-page application with responsive dashboards for all user roles.

02

API Gateway

RESTful endpoints handling authentication, data routing, and service orchestration.

03

Business Logic

Spring Boot services implementing core functionality and AI optimization algorithms.

04

Data Layer

PostgreSQL database storing fleet data, bookings, telemetry, and maintenance records.

System Workflow



Secure Login

Users authenticate and access role-specific dashboards with tailored functionality.



Booking Request

Customers select pickup and drop locations, triggering vehicle availability validation.



Data Processing

API calls update vehicle status and store booking information in real-time.



AI Optimization

Route optimization and maintenance prediction algorithms process available data.



Route Display

Optimal routes and vehicle tracking information appear on dashboards with live updates.

Authentication & Role Management

Secure access control built on industry-standard protocols with distinct permissions for each stakeholder role.

JWT Authentication

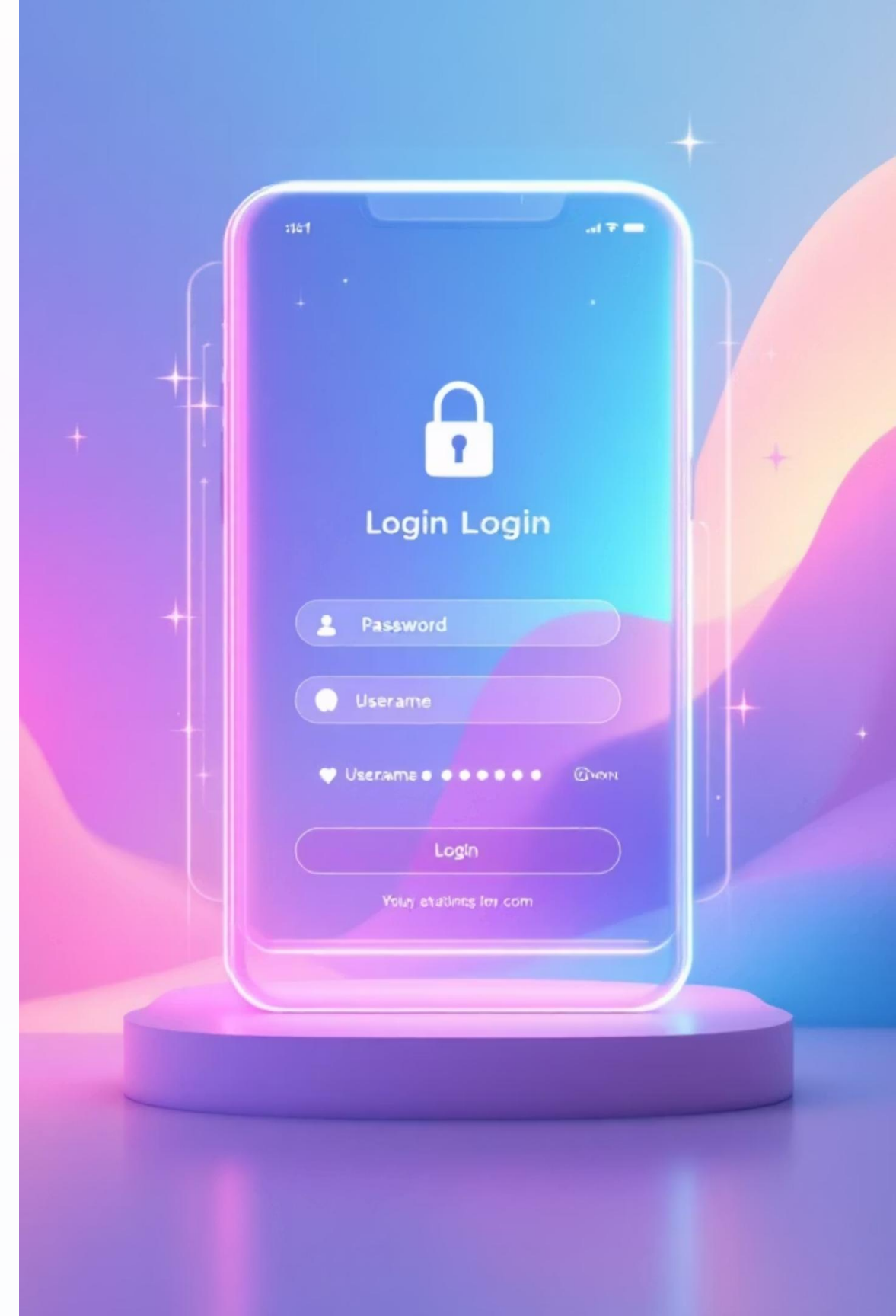
Token-based security ensuring encrypted, stateless authentication for all user sessions.

Role-Based Access

Three distinct roles: Customer for bookings, Driver for assigned trips, Admin for system monitoring.

Isolated Dashboards

Each role accesses separate dashboards and API endpoints with appropriate functionality.





Fleet Inventory & Vehicle Management

Real-time visibility into every vehicle in the fleet with automated status updates and comprehensive tracking capabilities.



Vehicle CRUD Operations

Full lifecycle management through Spring Boot APIs handling create, read, update, and delete operations.



Status Management

Automated state transitions between AVAILABLE, IN_USE, and AVAILABLE as bookings progress.



Telemetry Tracking

Continuous updates of location coordinates, fuel levels, speed, and operational parameters.

AI-Based Route Optimization

Intelligent navigation that analyzes multiple factors including distance, traffic patterns, and vehicle efficiency to deliver optimal route recommendations.

1

Fastest Route

Prioritizes minimum travel time by routing through low-traffic zones with real-time traffic avoidance.

2

Eco Route

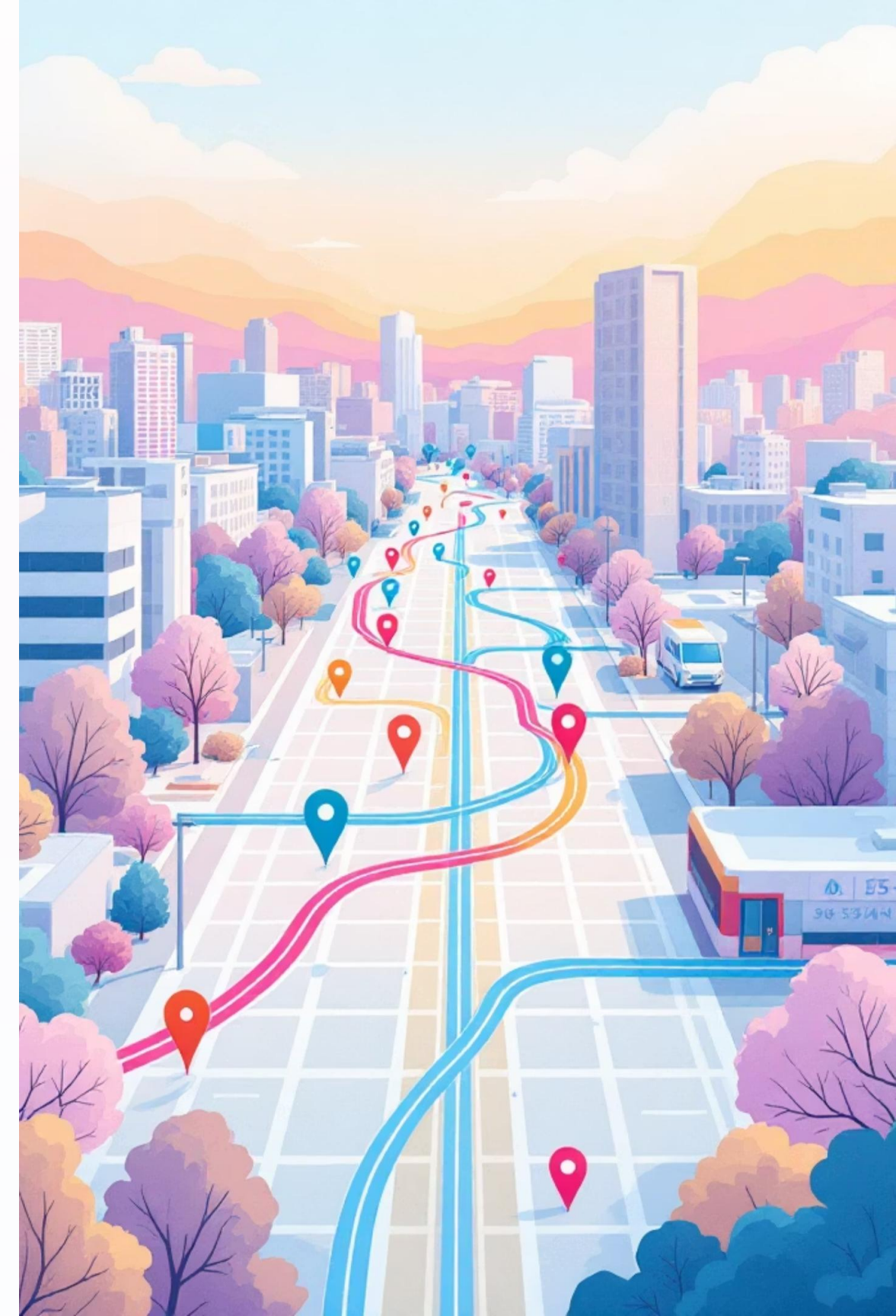
Optimizes fuel or battery efficiency, ideal for EV fleets by balancing speed with energy conservation.

3

Scenic Route

Alternate paths avoiding high-density traffic areas while providing reliable estimated arrival times.

Each route displays ETA, distance, estimated arrival time, and current traffic level with color-coded paths on simulated map views.



Predictive Maintenance & Real-Time Telemetry

Proactive maintenance alerts generated through AI analysis of real-time telemetry data including battery levels, engine health, tire pressure, and brake wear.



Battery Health

Capacity degradation tracking with charge cycle analysis for electric vehicle fleets.



Tire Monitoring

Pressure and wear analysis predicting replacement needs before safety thresholds are breached.



Engine Analysis

Performance metrics and anomaly detection identifying mechanical issues before failure occurs.



Brake Systems

Wear pattern analysis with predictive alerts ensuring safety compliance and preventing accidents.



Technology Stack & Results

Built with modern, enterprise-grade technologies delivering scalable performance and robust functionality.

Front-End

- *React with CDN integration*
- *HTML/CSS for SPA structure*
- *Leaflet.js for mapping*
- *OSRM API for routing*

Back-End

- *Spring Boot for APIs*
- *PostgreSQL for data*
- *RESTful architecture*
- *JWT for security*

3

Route Options

*Fastest, Eco, and Scenic
paths for every trip*

4

Vehicle Metrics

*Real-time telemetry
monitoring*

3

User Roles

*Customer, Driver, and
Admin dashboards*



Screenshots

